

ITSimplera Institute



Week 1 Task Report

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Networking Fundamentals & Network Simulation Tools

Local Area Network

Introduction

Networking is basically how devices talk to each other, letting them share data and resources. It's one of the core pieces of any IT system and something almost every organization relies on.

This report covers the work I completed in Week 1 of the Network Administration Internship Program at ITSimplera Institute. My main goal this week was to build a solid understanding of basic networking concepts, including network devices, IP addressing, and communication models.

To get some hands-on practice, I explored two widely used simulation tools: Cisco Packet Tracer and GNS3. Both are used by network professionals to design and test networks in a virtual environment before setting anything up on real hardware.

I built a basic LAN topology using both tools and assigned static IP addresses to the devices. I then used the ping command to test and confirm connectivity between them. This report walks through the steps I followed, the concepts I understood, and the difficulties I ran into along the way.

Networking Concepts Learned

Network: a group of devices connected together so they can share data and resources.

LAN (Local Area Network): connects devices within a small area, such as an office or a lab, and is used for fast, local communication.

IP Address: a unique number assigned to each device on a network, allowing devices to identify and communicate with each other.

Static IP Addressing: an IP address that is manually assigned to a device and doesn't change unless it's updated manually.

Network Devices:

- Router: connects different networks and directs data between them.
- Switch: connects multiple devices within the same network and forwards data to the correct device.
- PC/End Device: sends and receives data on the network.

Communication Models: networking follows standard models like OSI and TCP/IP, which divide network communication into layers, with each layer playing a specific role in sending and receiving data.

Ping Command: used to test connectivity between two devices by sending small packets of data and checking whether a response comes back.

Cisco Packet Tracer Interface Overview

Cisco Packet Tracer is a network simulation tool built by Cisco that lets you design, build, and test network topologies in a virtual environment.

Main Parts of the Interface:

Device Panel: found at the bottom left, holding categories of network devices like routers, switches, and end devices.

Workspace: the large area in the center where devices get dragged in to build the topology.

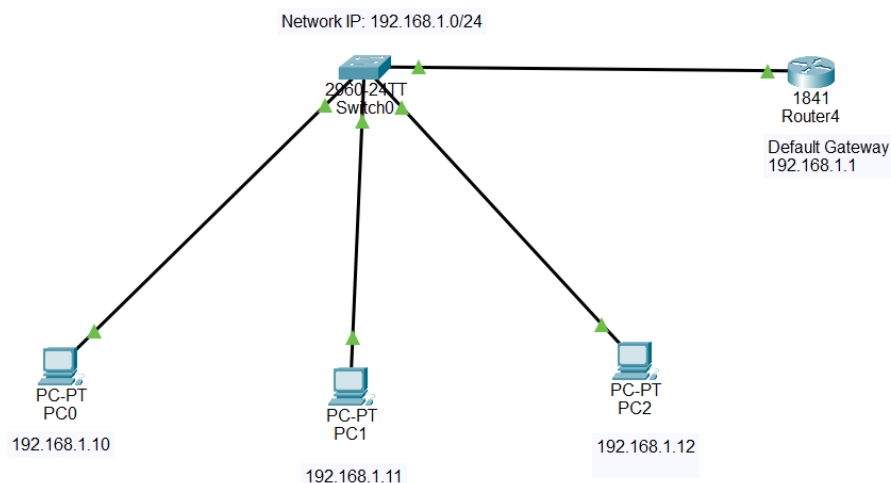
Connections Panel: holds the different cable types used to connect devices, such as copper straight-through and console cables.

Physical/Logical Toggle: lets you switch between a logical view of the network and a physical view.

Simulation Mode: lets you watch how data packets travel through the network step by step.

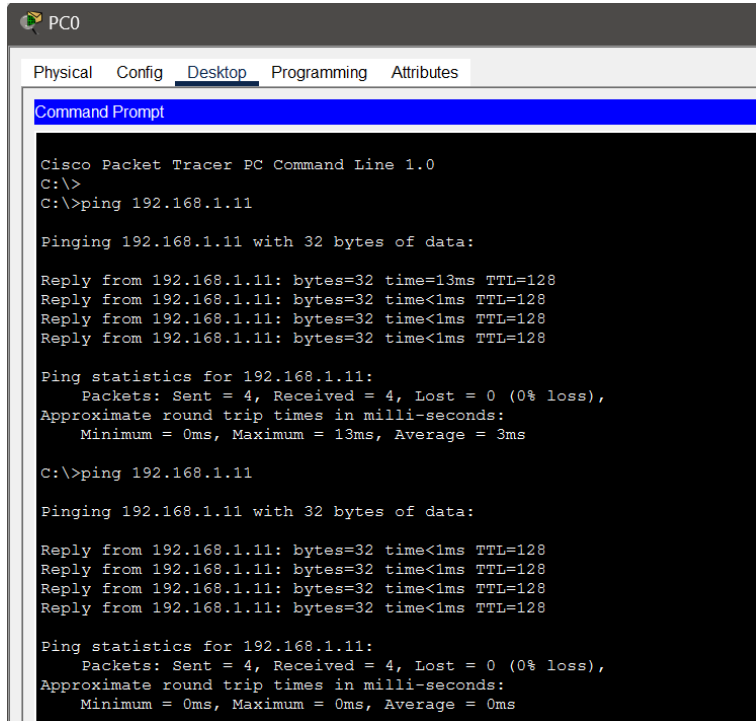
Command Line Interface (CLI): available inside routers and switches, used to configure devices with commands.

Packet Tracer Topology



Ping Test Results (Cisco Packet Tracer)

PC0: IP 192.168.1.10, used to ping PC1 (192.168.1.11) and PC2 (192.168.1.12)



```
PC0
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>
C:\>ping 192.168.1.11

Pinging 192.168.1.11 with 32 bytes of data:

Reply from 192.168.1.11: bytes=32 time=13ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 13ms, Average = 3ms

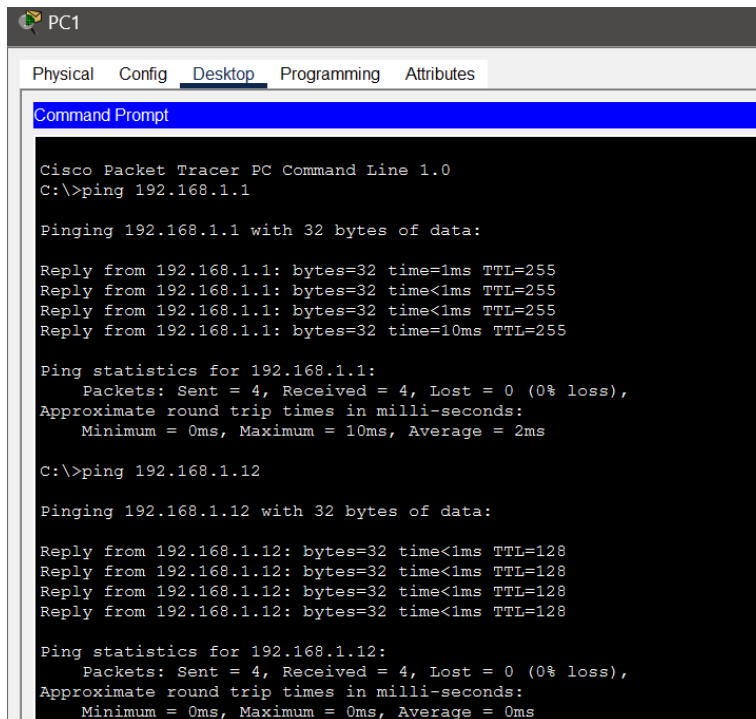
C:\>ping 192.168.1.11

Pinging 192.168.1.11 with 32 bytes of data:

Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

PC1: IP 192.168.1.11, used to ping the Default Gateway (192.168.1.1) and PC2 (192.168.1.12)



```
PC1
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=10ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 2ms

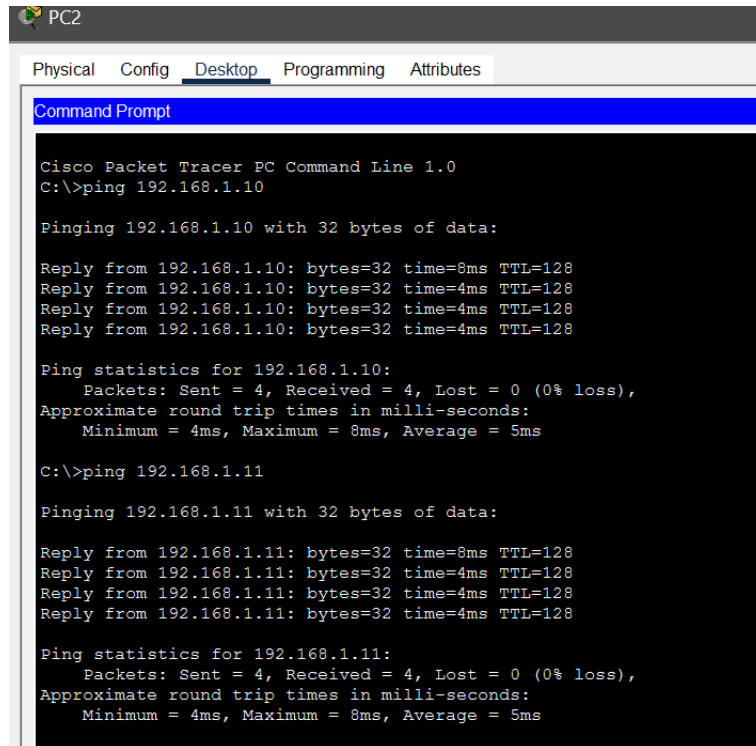
C:\>ping 192.168.1.12

Pinging 192.168.1.12 with 32 bytes of data:

Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

PC2: IP 192.168.1.12, used to ping PC0 (192.168.1.10) and PC1 (192.168.1.11)



```
PC2
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Reply from 192.168.1.10: bytes=32 time=8ms TTL=128
Reply from 192.168.1.10: bytes=32 time=4ms TTL=128
Reply from 192.168.1.10: bytes=32 time=4ms TTL=128
Reply from 192.168.1.10: bytes=32 time=4ms TTL=128

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 4ms, Maximum = 8ms, Average = 5ms

C:\>ping 192.168.1.11

Pinging 192.168.1.11 with 32 bytes of data:

Reply from 192.168.1.11: bytes=32 time=8ms TTL=128
Reply from 192.168.1.11: bytes=32 time=4ms TTL=128
Reply from 192.168.1.11: bytes=32 time=4ms TTL=128
Reply from 192.168.1.11: bytes=32 time=4ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 4ms, Maximum = 8ms, Average = 5ms
```

GNS3 Interface Overview

GNS3 (Graphical Network Simulator-3) is a network simulation tool used to design and test real network configurations using virtual and emulated devices.

Main Parts of the Interface:

Devices Toolbar: found on the left side, holding routers, switches, and end devices that can be added to the project.

Workspace: the central area where devices are placed and connected to build the topology.

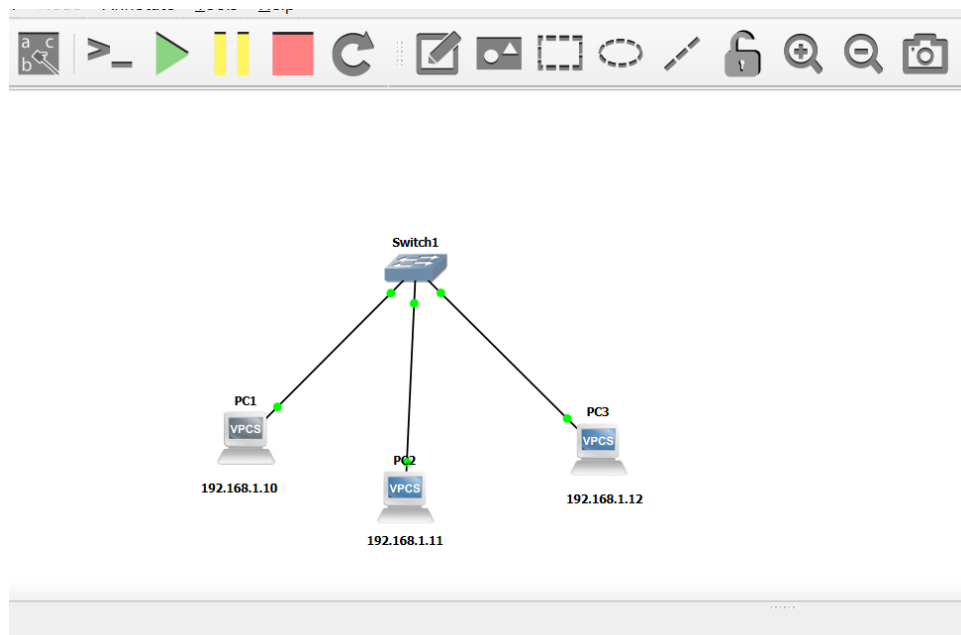
Topology Summary: shows a list of all devices added to the current project along with their status.

Console Access: devices are configured through a console connection, usually opened using PuTTY.

Server Summary: shows information about the local or remote GNS3 server running the simulation.

Start/Stop Controls: used to start, stop, or restart devices within the project.

GNS3 Topology



Ping Test Results (GNS3)

PC1: IP 192.168.1.10, used to ping PC2 (192.168.1.11) and PC3 (192.168.1.12)

```
PC1 - PuTTY

PC1> ping 192.168.1.11
84 bytes from 192.168.1.11 icmp_seq=1 ttl=64 time=1.959 ms
84 bytes from 192.168.1.11 icmp_seq=2 ttl=64 time=0.849 ms
84 bytes from 192.168.1.11 icmp_seq=3 ttl=64 time=1.866 ms
84 bytes from 192.168.1.11 icmp_seq=4 ttl=64 time=1.441 ms
84 bytes from 192.168.1.11 icmp_seq=5 ttl=64 time=0.425 ms

PC1> ping 192.168.1.12
84 bytes from 192.168.1.12 icmp_seq=1 ttl=64 time=0.681 ms
84 bytes from 192.168.1.12 icmp_seq=2 ttl=64 time=1.026 ms
84 bytes from 192.168.1.12 icmp_seq=3 ttl=64 time=1.008 ms
84 bytes from 192.168.1.12 icmp_seq=4 ttl=64 time=0.752 ms
84 bytes from 192.168.1.12 icmp_seq=5 ttl=64 time=1.406 ms

PC1>
```

PC2: IP 192.168.1.11, used to ping PC1 (192.168.1.10) and PC3 (192.168.1.12)

```
PC2 - PuTTY
PC2> ping 192.168.1.10
84 bytes from 192.168.1.10 icmp_seq=1 ttl=64 time=1.797 ms
84 bytes from 192.168.1.10 icmp_seq=2 ttl=64 time=2.238 ms
84 bytes from 192.168.1.10 icmp_seq=3 ttl=64 time=2.743 ms
84 bytes from 192.168.1.10 icmp_seq=4 ttl=64 time=2.346 ms
84 bytes from 192.168.1.10 icmp_seq=5 ttl=64 time=1.718 ms

PC2> ping 192.168.1.12
84 bytes from 192.168.1.12 icmp_seq=1 ttl=64 time=1.908 ms
84 bytes from 192.168.1.12 icmp_seq=2 ttl=64 time=1.131 ms
84 bytes from 192.168.1.12 icmp_seq=3 ttl=64 time=1.442 ms
84 bytes from 192.168.1.12 icmp_seq=4 ttl=64 time=1.929 ms
84 bytes from 192.168.1.12 icmp_seq=5 ttl=64 time=1.635 ms
```

PC3: IP 192.168.1.12, used to ping PC1 (192.168.1.10) and PC2 (192.168.1.11)

```
PC3> ip 192.168.1.12 255.255.255.0
Checking for duplicate address...
PC1 : 192.168.1.12 255.255.255.0

PC3> ping
PC3> ping 192.168.1.10
84 bytes from 192.168.1.10 icmp_seq=1 ttl=64 time=0.729 ms
84 bytes from 192.168.1.10 icmp_seq=2 ttl=64 time=1.293 ms
84 bytes from 192.168.1.10 icmp_seq=3 ttl=64 time=1.100 ms
84 bytes from 192.168.1.10 icmp_seq=4 ttl=64 time=0.966 ms
84 bytes from 192.168.1.10 icmp_seq=5 ttl=64 time=0.850 ms

PC3> ping 192.168.1.11
84 bytes from 192.168.1.11 icmp_seq=1 ttl=64 time=1.815 ms
84 bytes from 192.168.1.11 icmp_seq=2 ttl=64 time=1.052 ms
84 bytes from 192.168.1.11 icmp_seq=3 ttl=64 time=0.753 ms
84 bytes from 192.168.1.11 icmp_seq=4 ttl=64 time=1.148 ms
84 bytes from 192.168.1.11 icmp_seq=5 ttl=64 time=1.106 ms
```

Challenges Faced

When I opened GNS3 for the first time, the Routers panel was empty. Unlike Packet Tracer, GNS3 doesn't come with built-in router images, so a router appliance or IOS image needs to be added manually before it shows up.

I tried adding a Cisco 3725 router, but it failed because the required IOS image file wasn't available on my system. Cisco IOS images are licensed software, so they can't just be downloaded freely, which is what caused this issue.

A few of the other GNS3 router options I looked at, like the Qemu-based ones, also needed the GNS3 VM to be set up with VirtualBox or VMware before they could run, which was an extra setup step I hadn't expected coming from Packet Tracer. Because of this, I completed the GNS3 topology at the switch and PC level for now, and I plan to add the router once I have access to a licensed IOS image or a free alternative.

Conclusion

This week helped me build a solid foundation in basic networking concepts, including IP addressing, network devices, and communication models. I got hands-on experience with both Cisco Packet Tracer and GNS3 by building a simple LAN topology in each tool, assigning static IP addresses, and verifying connectivity through ping tests. Working through the router setup challenges in GNS3 also gave me a clearer picture of the practical, real-world considerations involved in network simulation. Overall, this task strengthened my practical networking skills and gave me a strong base for the more advanced tasks ahead.